



Event-Based Visual Encoding for Trustworthy Predictive Maintenance

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Context, Problem, and Goal:

Can we trust Visual Streams in the Industrial Internet of Things?

OBD & Monitoring



Current Status

Big Rock Detected

Feeder	Hopper Upper	Hopper Lower
No issue detected	Big rock present	No issue detected

Get real-time video feeds

Ripik.AI

!!ALERT!!

BIG ROCK DETECTED

Limestone feeding Report from~
2024-11-08T12:10:13 to
2024-11-08T13:10:13 on
belt camJ31c2.

SIZING DETAILS:
0-10mm 25.76%
10-20mm 10.16%
20-30mm 16.90%
30-40mm 22.94%
40-50mm 13.70%

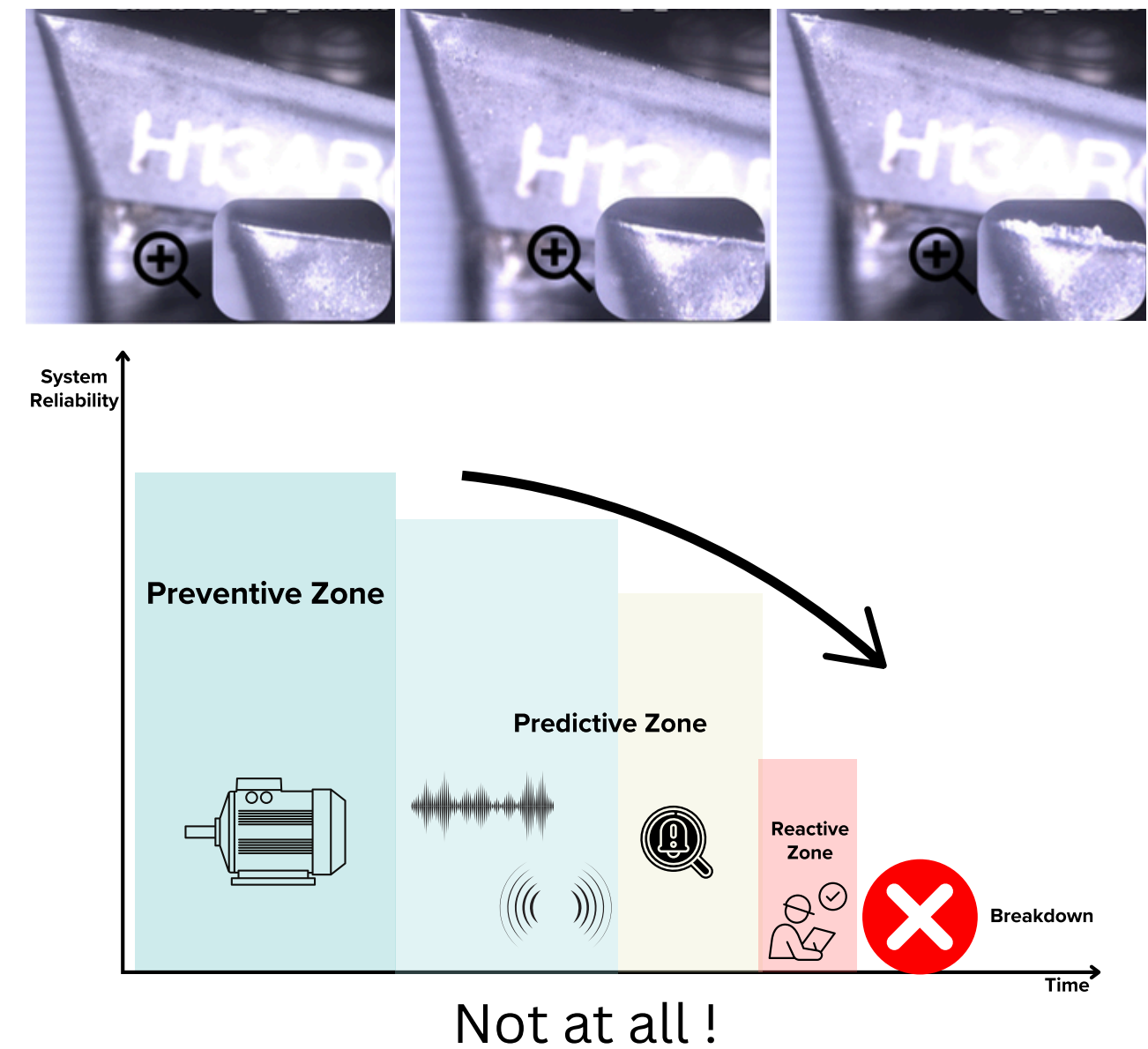
MOISTURE DETAILS:
14.29% (limit < 15%)

Regards,
Ripik AI

Real-time Alerts

Definetely !

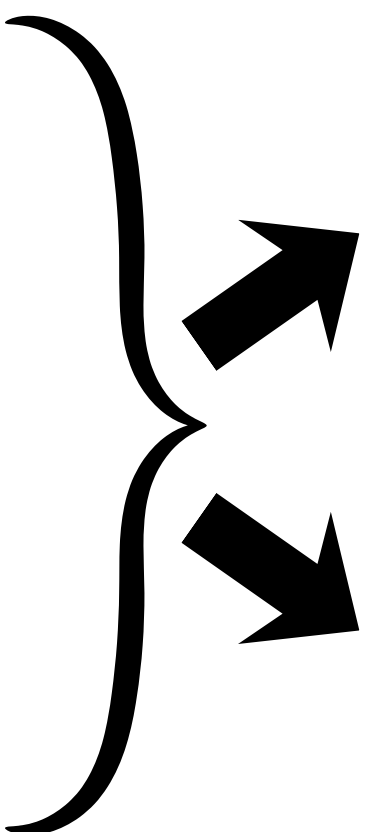
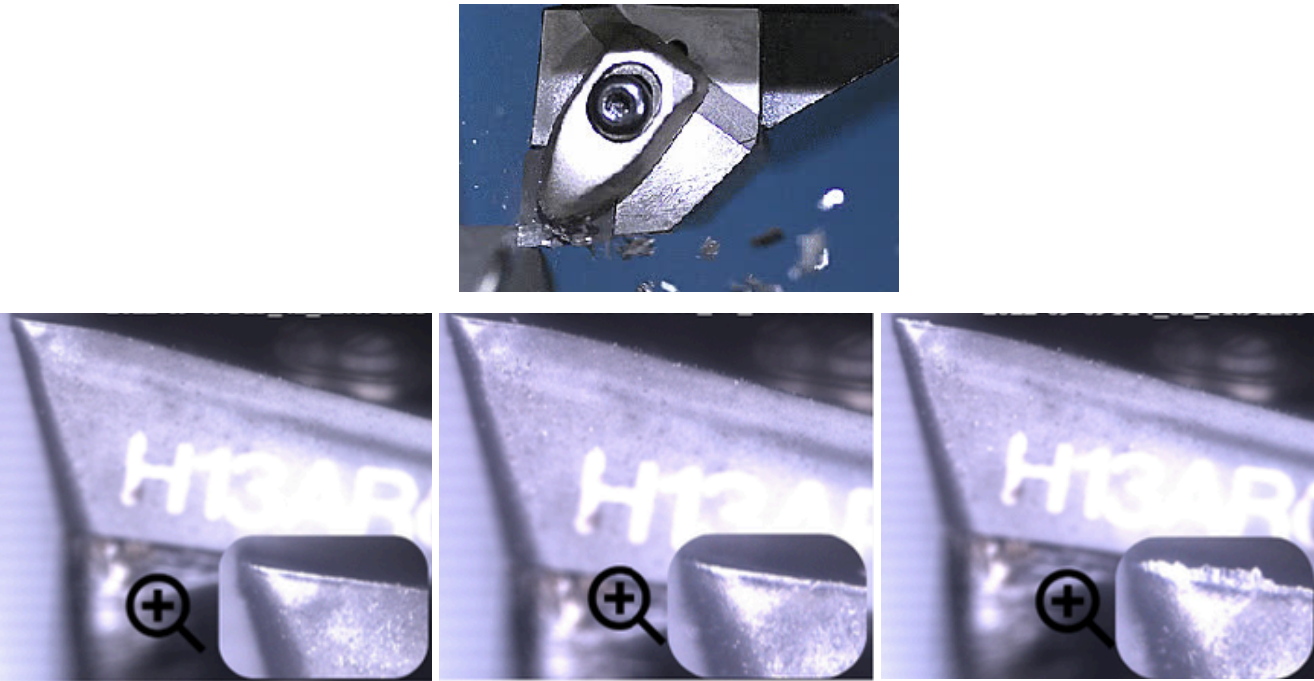
Pixel Level & Slow Degradation



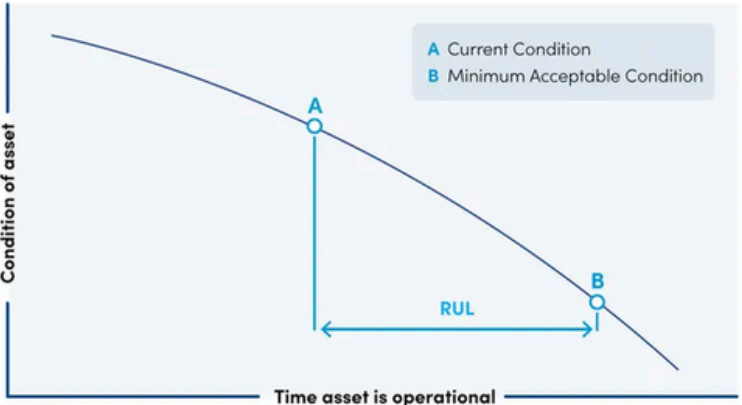
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Computer Vision in Predictive Maintenance

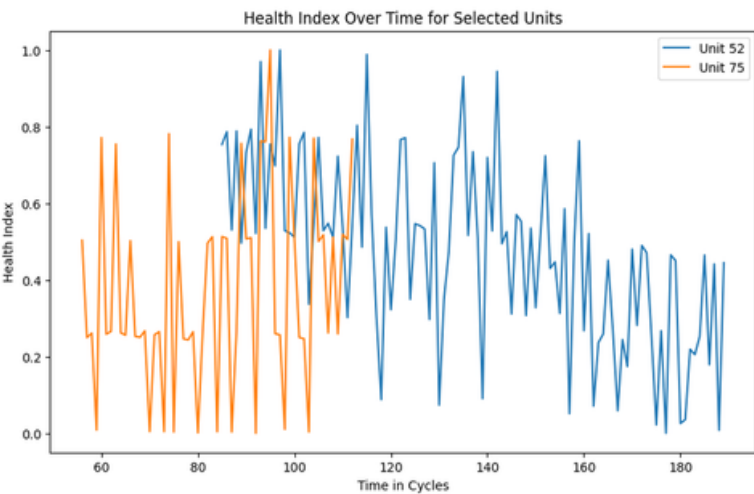
How can we use visual features?



Remaining Useful Life



Health Index



IDEAL

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Case Study: MATWI Dataset



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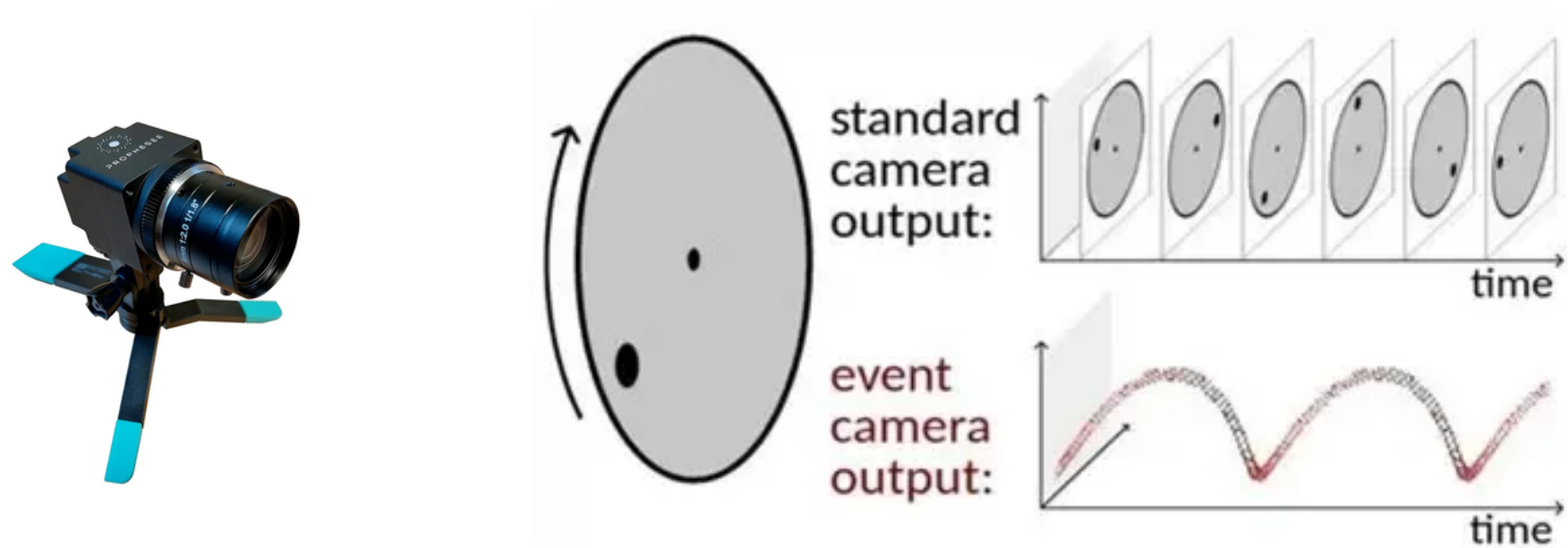
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Event Based Approach: An Overview

Event Based Technology cameras



Event Based Visual Encoding



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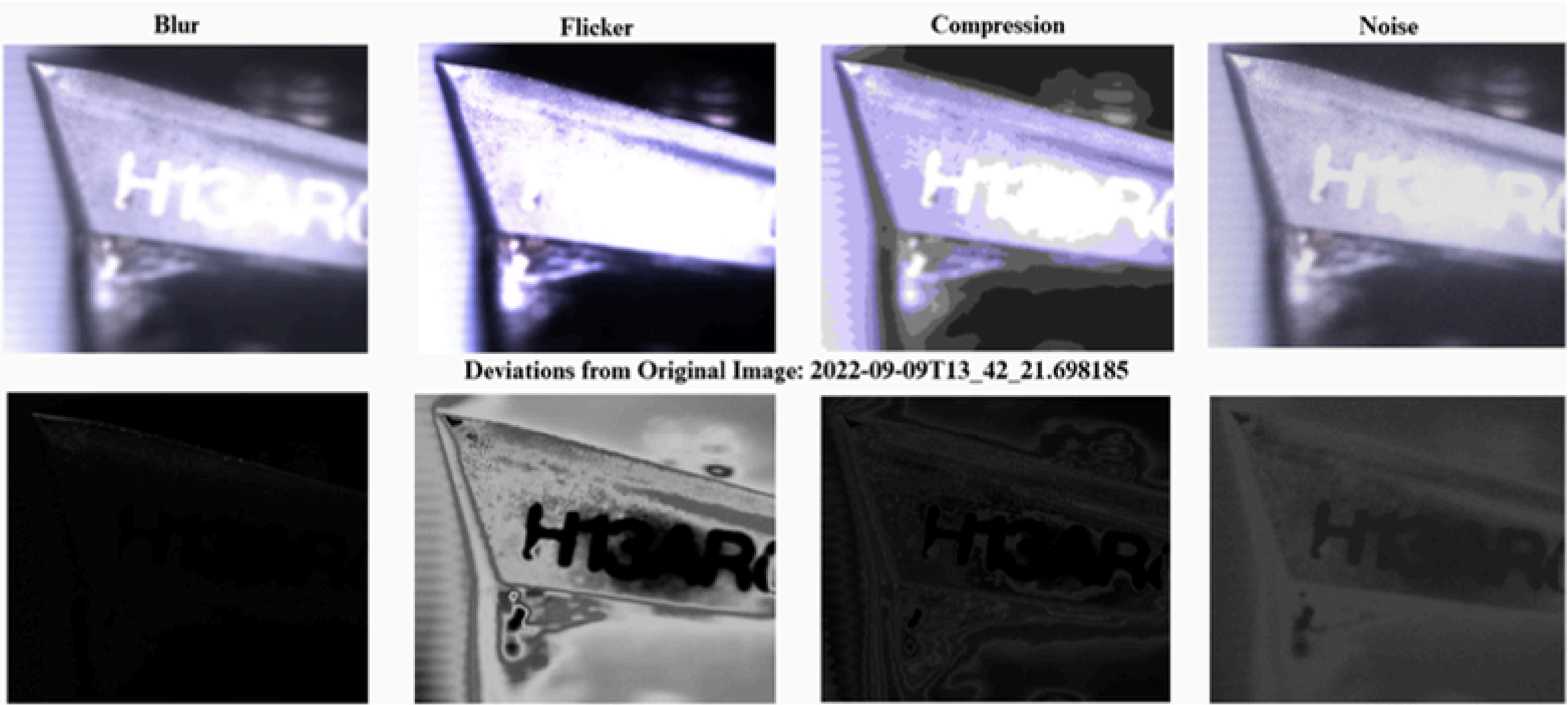
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Adversarial Conditions in Industrial Plants



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Event Based Visual Encoding for Industrial Predictive Maintenance

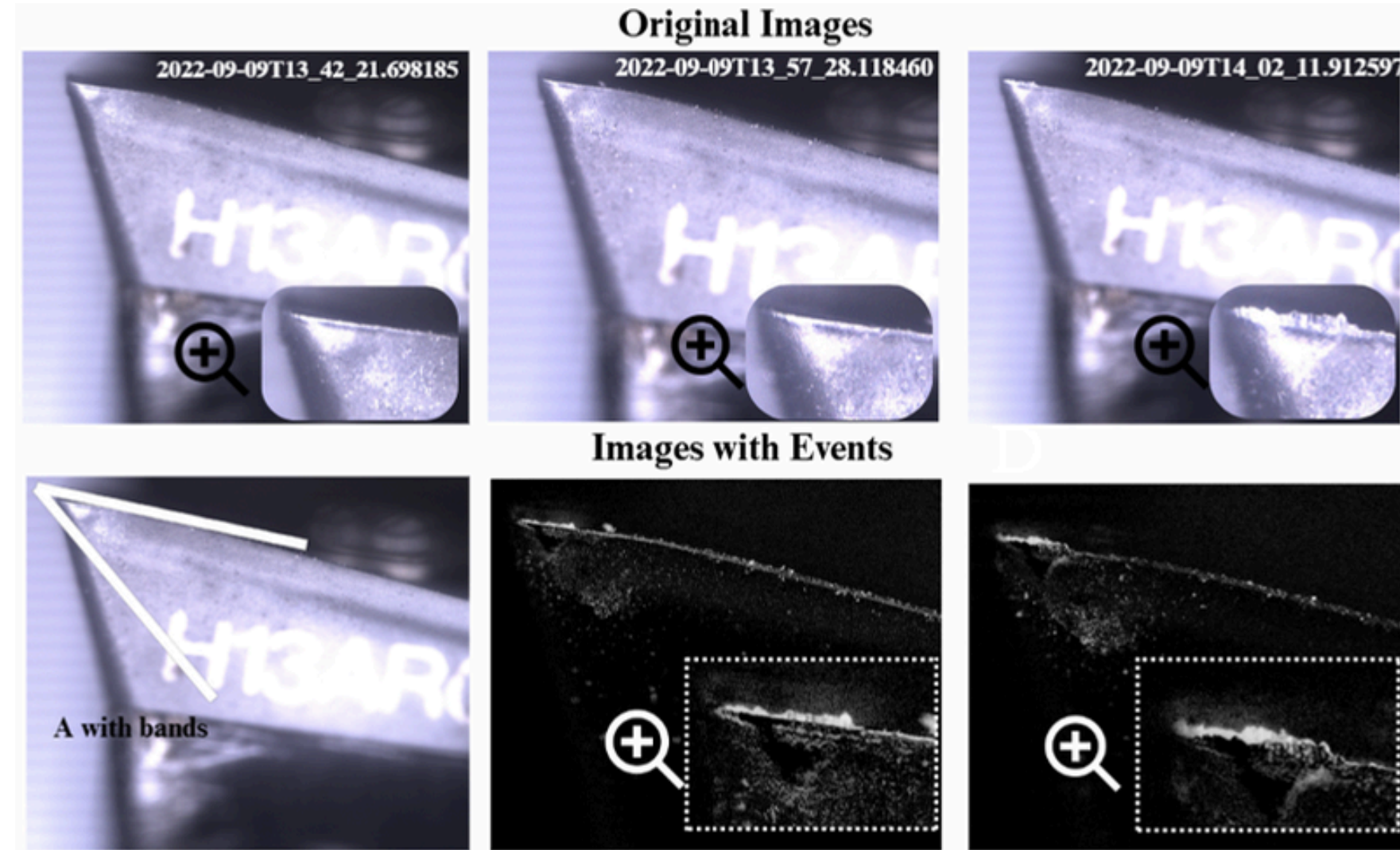


Table 1: Event descriptors from edge band

Descriptor	Definition (formula)
Band histogram pmf $p_{t,b}(u)$	$p_{t,b}(u) = \frac{1}{ \mathcal{B}_b } \sum_{x \in \mathcal{B}_b} \mathbb{1}\{A_t^\Sigma(x) = u\}, \quad u \in \{0, \dots, L\}, \quad b \in \{1, 2\}$ <i>Compactly summarizes band activity over count levels.</i>
Entropy $H_{t,b}$	$H_{t,b} = - \sum_{u=0}^L p_{t,b}(u) \log(p_{t,b}(u) + \varepsilon).$ <i>Dispersion of activity: localized vs. diffuse change.</i>
Dominant peak bin $u_{t,b}^*$	$u_{t,b}^* = \arg \max_{u \in \{0, \dots, L\}} p_{t,b}(u).$ <i>Most frequent activity level in the band.</i>
Peak width $W_{t,b}$	$W_{t,b} = \sum_{u=0}^L \mathbb{1}\{p_{t,b}(u) \geq \rho p_{t,b}(u_{t,b}^*)\}, \quad \rho \in (0, 1).$ <i>Spread around the dominant mode (broadens under change).</i>
Temporal consistency $C_{t,b}$	$C_{t,b} = \text{corr}(\text{vec}(A_t^\Sigma \mathcal{B}_b), \text{vec}(A_{t-1}^\Sigma \mathcal{B}_b)).$ <i>Persistence vs. transient noise/artifacts.</i>
Inter-band contrast D_t	$D_t = u_{t,1}^* - u_{t,2}^* \quad (\text{or } u_{t,1}^* - u_{t,2}^*).$ <i>Boundary-localized change via inside-outside comparison.</i>
Composite Descriptor $\mathbf{f}_t$	$\mathbf{f}_t = [H_{t,1}, H_{t,2}, W_{t,1}, W_{t,2}, C_{t,1}, C_{t,2}, D_t]^\top \in \mathbb{R}^7(7)$ $\varepsilon > 0$ avoids $\log(0)$; L is the event-aggregation length; ρ controls the peak-width threshold.



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Diagnostic & Trust Gate

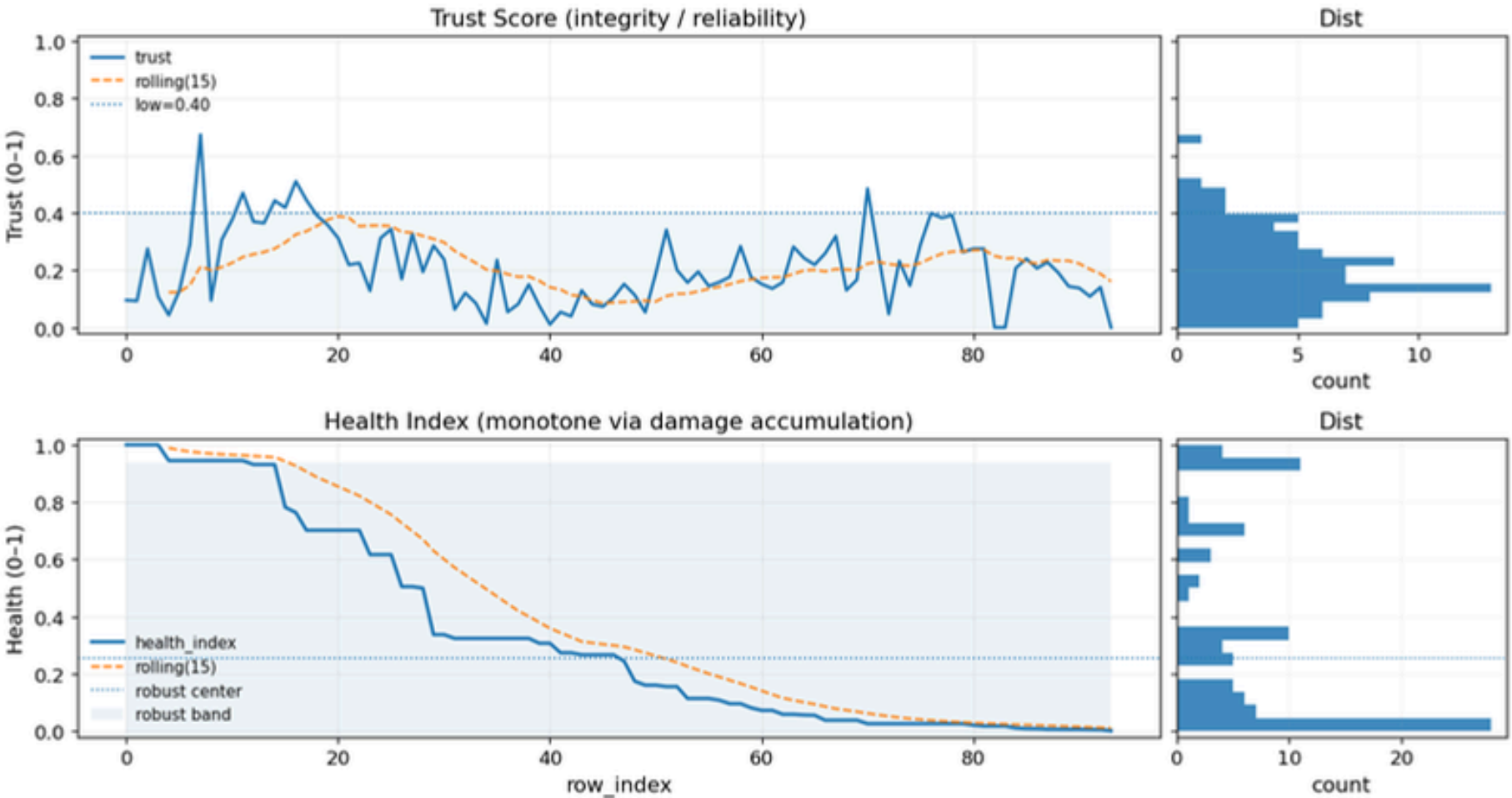


Table 2: HI construction with trust gating.

Component	Equation
<i>Robust HI score</i>	
Robust normalization	$\tilde{f}_{t,i} = \frac{f_{t,i} - \text{median}(f_{1:N_0,i})}{\text{MAD}(f_{1:N_0,i}) + \epsilon} \quad (8)$ <i>Early prefix baseline; robust and leakage-safe.</i>
Wear-score fusion	$s_t = \sum_{i=1}^d w_i \phi(\tilde{f}_{t,i}), \quad w_i \geq 0 \quad (9)$ <i>One wear proxy; no cancellation, outlier-bounded.</i>
Running range tracking	$m_t = \min(m_{t-1}, s_t), \quad M_t = \max(M_{t-1}, s_t), \quad (10)$ $r_t = M_t - m_t$ <i>Range expands only at new extremes of s_t.</i>
Monotone HI update	$\Delta r_t = \max(0, r_t - r_{t-1}),$ $h_t = \text{clip}_{[0,1]}(h_{t-1} - \alpha \Delta r_t), \quad h_1 = 1 \quad (11)$ <i>Decreases only on positive range expansions; guarantees monotonicity.</i>
<i>Trust score computation</i>	
Trust score (reliability)	$z_t^{(j)} = \frac{g_t^{(j)} - \text{median}(g_{1:N_0}^{(j)})}{\text{MAD}(g_{1:N_0}^{(j)}) + \epsilon}, \quad (12)$ $\rho_t = \sum_{j=1}^p z_t^{(j)} , \quad q_t = \exp(-\beta \rho_t)$ <i>Aggregates robust deviations of reliability</i>
Trust-gated update	$h_t = \text{clip}_{[0,1]}(h_{t-1} - \alpha q_t \Delta r_t) \quad (13)$ <i>Down-weights HI decrements under low reliability.</i>
Trust debt	$d_t = \lambda d_{t-1} + (1 - q_t) \Delta r_t, \quad \lambda \in [0, 1) \quad (14)$ <i>Tracks deferred damage under sustained low trust.</i>



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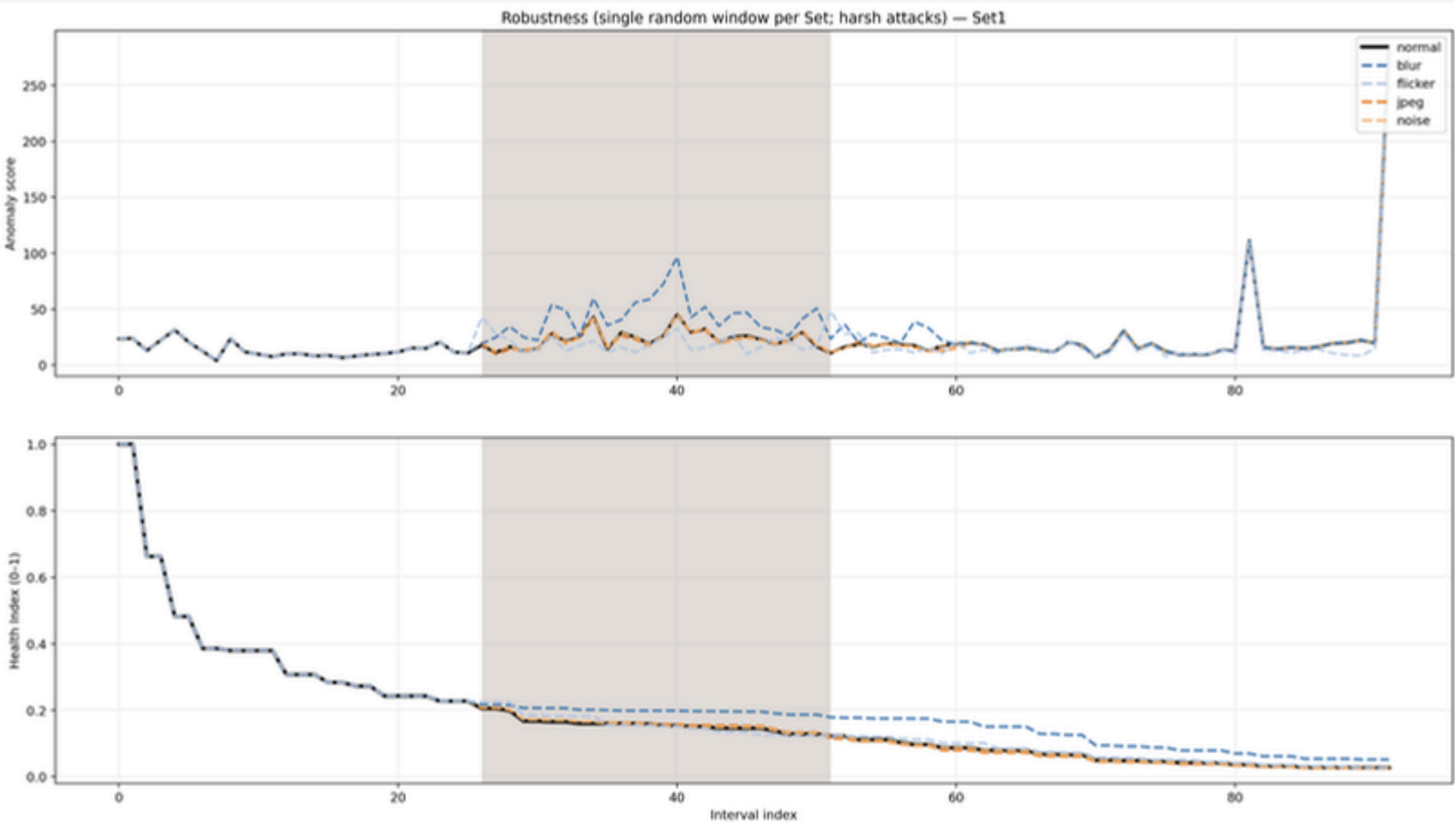
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Effects of Adversarial Condition



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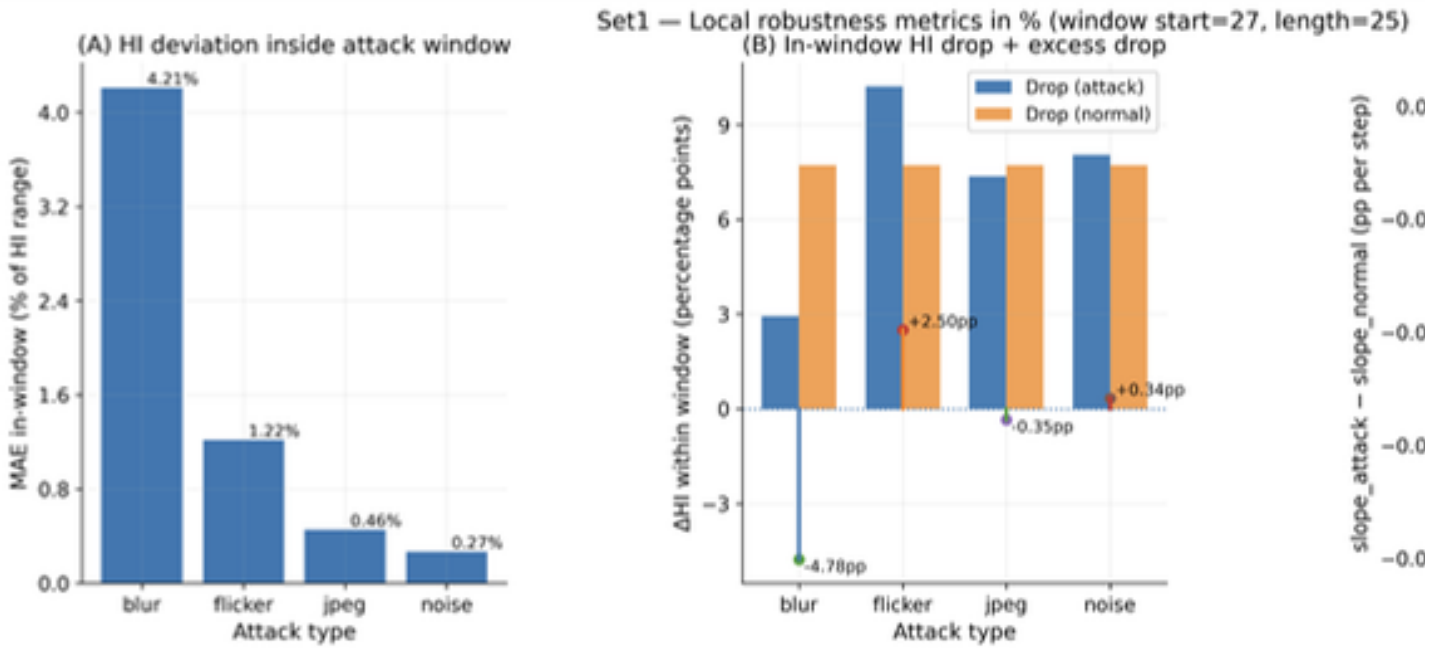
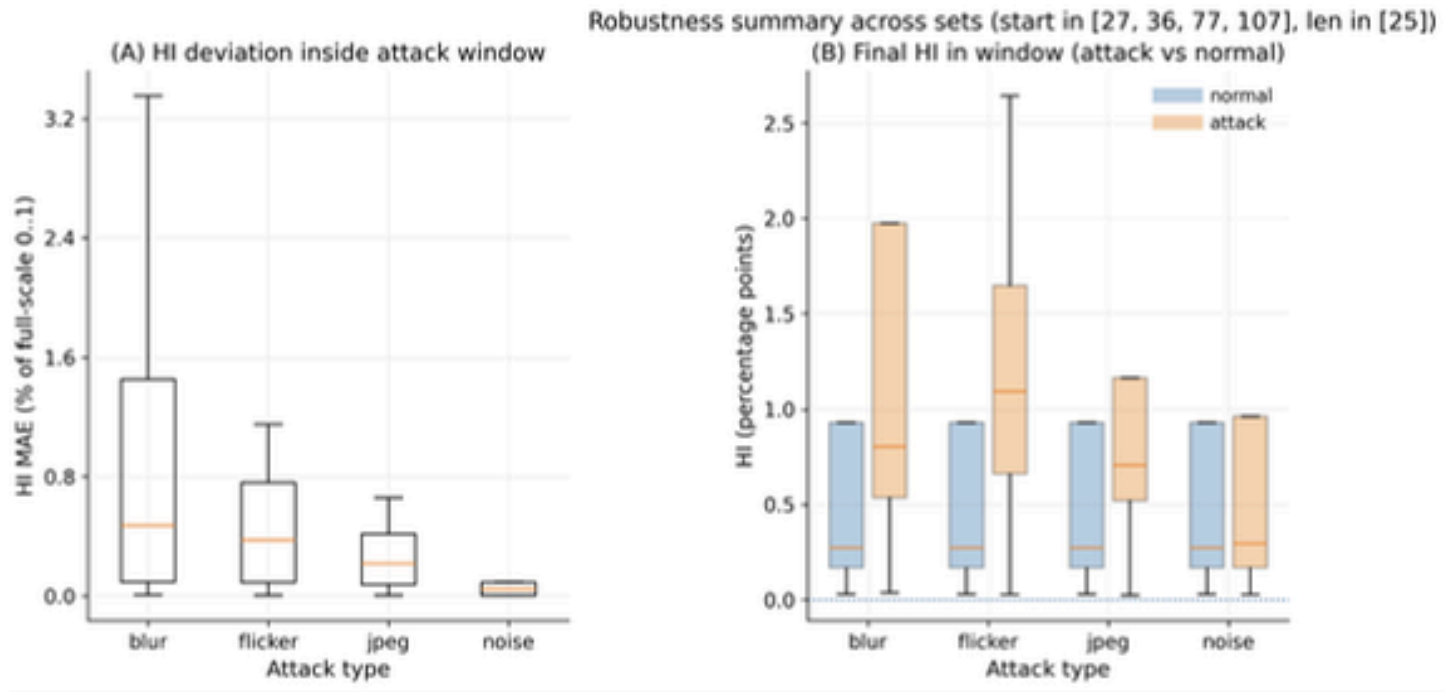
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EVE Local & Global Performances



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References

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Experiments on MATWI under severe single-window corruptions demonstrate stable HI behavior with minimal drift, supporting EVE representations as a robust interface between vision and PdM.

Current Limitations:

- physical interpretability of the aggregated descriptors.
- Static ROI band
- Blurring sensitivity

Future works:

- Integration with Predictive Maintenance Forecasting
- Dynamic and automatic ROI bands for flexible deployment
- Targeted adversarial manipulation under different threat conditions

Questions?



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